Python Activity Record

Advanced Numeric Pattern Generator

1. Problem Statement

Develop a Python program that accepts a diamond size from 1 to 20 and prints a hollow diamond pattern using asterisks (*). The program checks the entered size and displays an error message if the value is outside the permitted range.

2. Objectives

- To practice taking integer input from the user.
- To validate input using an if-else condition.
- To use loops and ranges to print the upper and lower halves of a diamond.
- To use spaces and asterisks to create a hollow pattern.
- To display a clear message when the size is invalid.

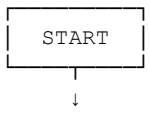
3. Assumptions

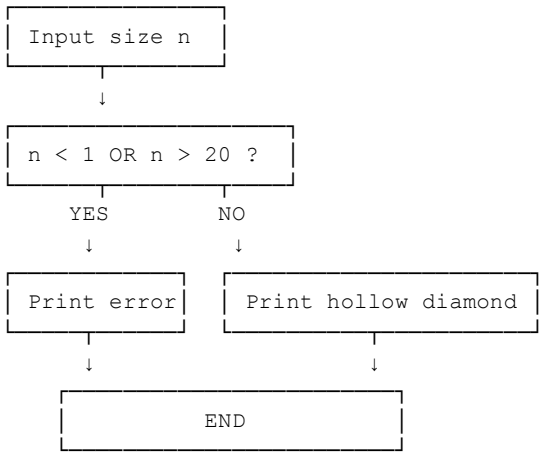
- The user enters an integer when prompted.
- A valid diamond size is from 1 to 20, including both limits.
- The pattern uses the asterisk (*) character.
- The program is run in a Python environment that supports input() and print().
- Non-integer input is not handled by the supplied code and may cause a ValueError.

4. Algorithm

1. Start the program.
2. Ask the user to enter the diamond size n.
3. Check whether n is less than 1 or greater than 20.
4. If invalid, print “Size must be from 1 to 20.”
5. Otherwise, create row numbers from 1 to n, followed by n−1 down to 1.
6. For each row, calculate the leading spaces as n−row.
7. If row equals 1, print the leading spaces followed by one asterisk.
8. Otherwise, print the leading spaces, an asterisk, the inner spaces, and another asterisk.
9. Stop the program.

5. Flowchart





6. Pseudocode

```
BEGIN
    INPUT n
    IF n < 1 OR n > 20 THEN
        PRINT "Size must be from 1 to 20."
    ELSE
        rows = numbers from 1 to n, then n-1 down to 1
        FOR each row IN rows
            spaces = n - row
            IF row == 1 THEN
                PRINT spaces + "*"
            ELSE
                inner_spaces = 2 * row - 3
                PRINT spaces + "*" + inner_spaces spaces + "*"
            END IF
        END FOR
    END IF
END
```

7. Python Source Code

```
n = int(input("Diamond size (1-20): "))

if n < 1 or n > 20:

    print("Size must be from 1 to 20.")

else:

    for row in list(range(1, n + 1)) + list(range(n - 1, 0, -1)):

        spaces = " " * (n - row)

        if row == 1:

            print(spaces + "*")
```

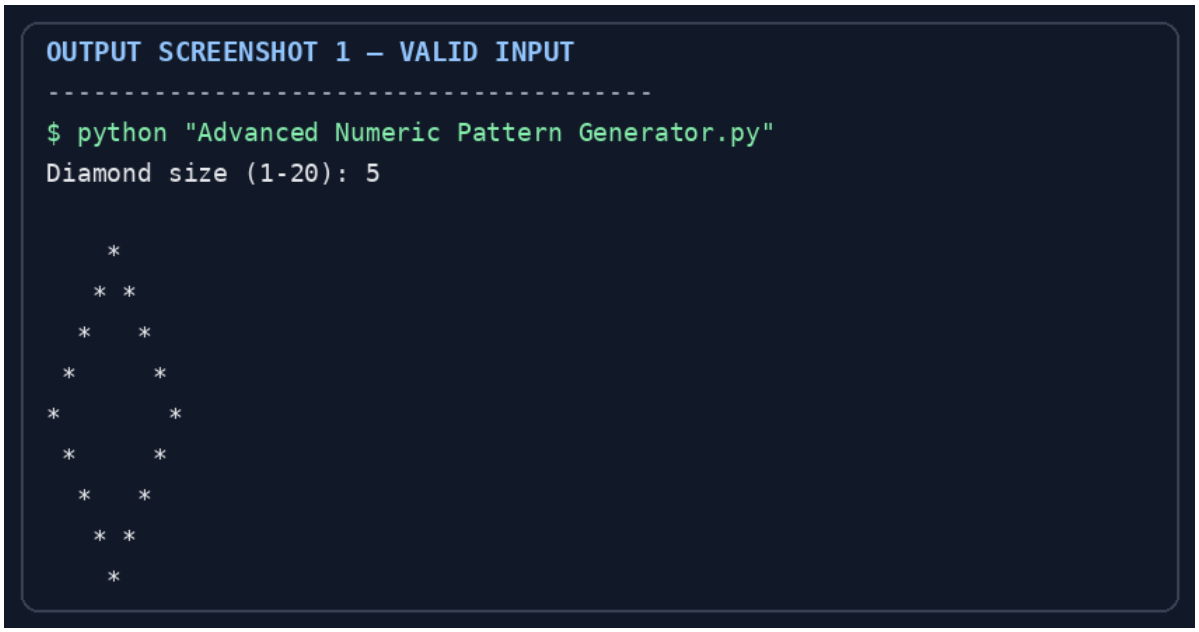
```
        else:
            print(spaces + "*" + " " * (2 * row - 3) + "*")
```

8. Test Cases

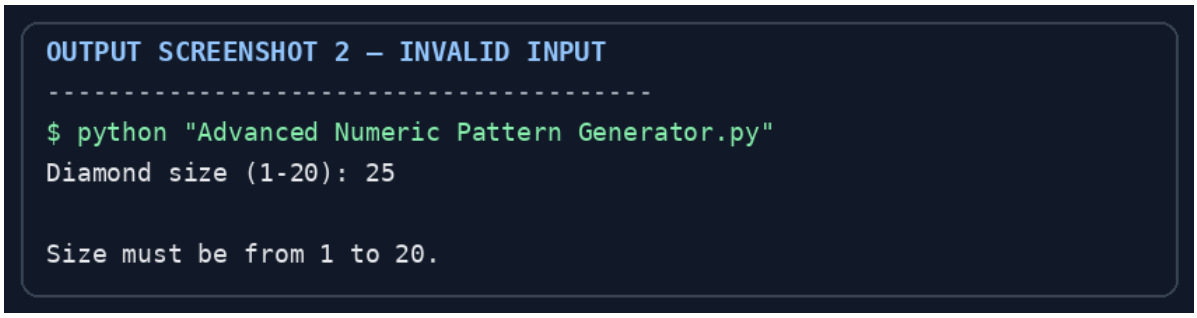
Test	Input	Expected Output	Purpose
1	1	One asterisk (*)	Minimum valid size
2	5	Hollow diamond with 9 rows	Normal valid input
3	20	Hollow diamond with 39 rows	Maximum valid size
4	0	Size must be from 1 to 20.	Below minimum
5	25	Size must be from 1 to 20.	Above maximum

9. Output Screenshots

Screenshot 1: Valid input (n = 5).



Screenshot 2: Invalid input (n = 25).



10. Conclusion

The Advanced Numeric Pattern Generator demonstrates input validation, conditional statements, loops, ranges, and string operations in Python. For valid sizes, it prints a hollow diamond pattern. For values outside the range 1 to 20, it displays an error message. The program works as intended when the user enters an integer.

Limitation: The supplied code does not validate non-integer input such as letters; this can raise a ValueError.